Chapter 10: MAP Estimation, Information Theory, and Markov Chains

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CSAI2121 Probability and Statistics
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Learning Objectives

By the end of this lecture, you should be able to:

- Write Bayes' theorem for an unknown parameter
- Define and compute a **Maximum A Posteriori** (MAP) estimate
- Explain how a Gaussian prior leads to **L2 regularization**
- Explain how a Laplace prior leads to **L1 regularization**
- Compute self-information, entropy, and information gain
- Compute cross-entropy and KL divergence, and explain their relationship
- Explain why cross-entropy is used as a classification loss
- Describe a Markov Chain using a transition matrix [Not Examined]

Roadmap of Today

Part	Main topics	Time
A. MAP Estimation	Prior, posterior, regularization	90 min
B. Information Theory	Entropy, cross-entropy, KL	60 min
C. Markov Chains	States, transition matrix	30 min

Assessment Scope

- Parts A and B are examinable.
- Part C is **not examined** in the Test or Examination; it is assessed only through a short classwork.
- Slides marked **[Extension]** are for further reading only.

Maximum A Posteriori Estimation

Motivation: When MLE Is Too Extreme

Recall Chapter 8: a coin is flipped 3 times and gives

$$H,\ H,\ H.$$

The MLE is:

$$\widehat{p}_{\mathrm{MLE}} = \frac{k}{n} = \frac{3}{3} = 1.$$

- This claims that tails is **impossible**, which is unreasonable after only 3 flips.
- In Chapter 9, a model with many features fitted to few data can also produce extreme, unstable weights.

Question

How can we combine the observed data with sensible **prior knowledge** about the parameter?

Answer

Treat the parameter as uncertain and use Bayes' theorem.

Bayes' Theorem for Parameters

In Week 2, Bayes' theorem was applied to events. Now we apply it to a parameter θ and data D:

$$\boxed{p(\theta \mid D) = \frac{p(D \mid \theta)\, p(\theta)}{p(D)}}$$

Term	Name	Meaning
$p(\theta)$	Prior	Belief about θ before seeing data
$p(D \mid \theta)$	Likelihood	How well θ explains the data
$p(\theta \mid D)$	Posterior	Updated belief after seeing data
$p(D)$	Evidence	Normalizing constant

Since $p(D)$ does not depend on θ:

$$\text{posterior} \;\propto\; \text{likelihood} \times \text{prior}.$$

The MAP Estimate

Definition

The Maximum A Posteriori estimate is the **mode** of the posterior:

$$\boxed{\widehat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} p(\theta \mid D) = \arg\max_{\theta} p(D \mid \theta)\, p(\theta)}$$

Taking logarithms, as for MLE:

$$\widehat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \Big[\underbrace{\log p(D \mid \theta)}_{\ell(\theta)} + \log p(\theta) \Big].$$

Key Point

MAP = log-likelihood + log-prior. If the prior is constant (uniform), the MAP estimate is the same as the MLE.

MLE vs. MAP

	MLE	**MAP**
Objective	$p(D \mid \theta)$	$p(D \mid \theta)\, p(\theta)$
Uses a prior?	No	Yes
Small samples	May be extreme	Pulled towards the prior
Large samples	Data dominate	Data dominate; MAP $\approx$ MLE
ML counterpart	Unregularized loss	Loss + regularization

Common Misunderstanding

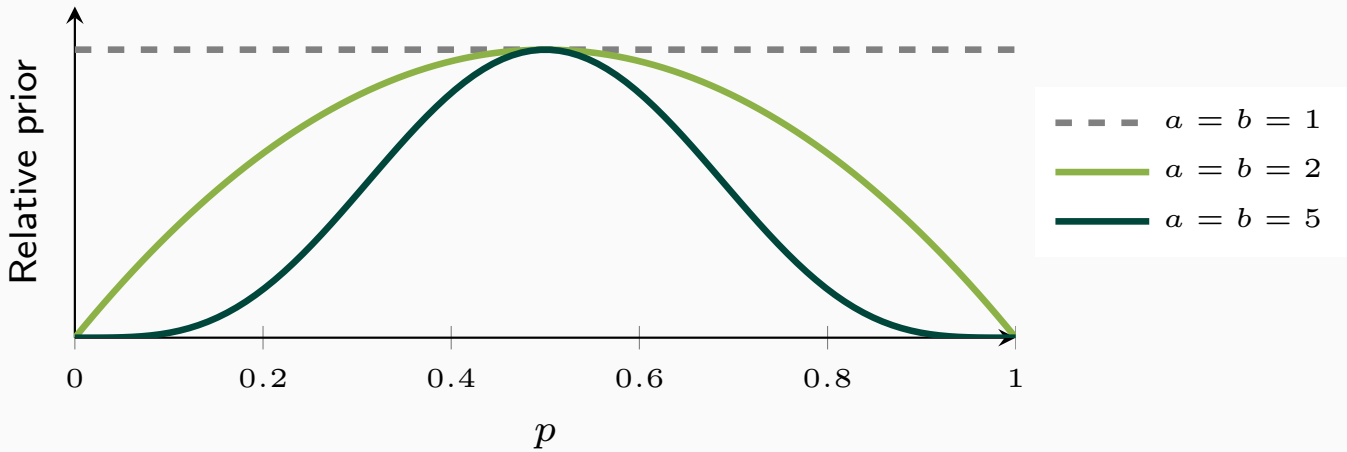
MAP is still a **single point estimate**. A full Bayesian analysis keeps the entire posterior distribution; MAP keeps only its peak.

A Prior for a Probability Parameter

For $0 < p < 1$, a convenient family of priors is:

$$p(p) \;\propto\; p^{a-1}(1-p)^{b-1}, \qquad a, b > 0.$$

(This is the *Beta*(a, b) distribution; only its shape matters here.)



- $a = b = 1$: uniform prior (no preference).
- Larger $a = b$: a stronger belief that p is near 0.5.

Deriving the Bernoulli MAP Estimate

With k successes in n trials:

$$p(p \mid D) \;\propto\; \underbrace{p^k(1-p)^{n-k}}_{\text{likelihood}} \times \underbrace{p^{a-1}(1-p)^{b-1}}_{\text{prior}} = p^{k+a-1}(1-p)^{n-k+b-1}.$$

Log-posterior (up to a constant):

$$(k+a-1)\log p + (n-k+b-1)\log(1-p).$$

This has exactly the same form as the Bernoulli log-likelihood in Chapter 8, with $k \to k+a-1$ and $n \to n+a+b-2$. Hence:

$$\boxed{\widehat{p}_{\mathrm{MAP}} = \frac{k+a-1}{n+a+b-2}}$$

Check: $a = b = 1$ gives $\widehat{p}_{\mathrm{MAP}} = k/n = \widehat{p}_{\mathrm{MLE}}$.

Worked Example: Three Heads in Three Flips

Data: $n = 3$, $k = 3$. Prior: $a = b = 2$.

$$\widehat{p}_{\mathrm{MLE}} = \frac{3}{3} = 1, \qquad \widehat{p}_{\mathrm{MAP}} = \frac{3 + 2 - 1}{3 + 2 + 2 - 2} = \frac{4}{5} = 0.8.$$

Pseudo-count Interpretation

With $a = b = 2$:

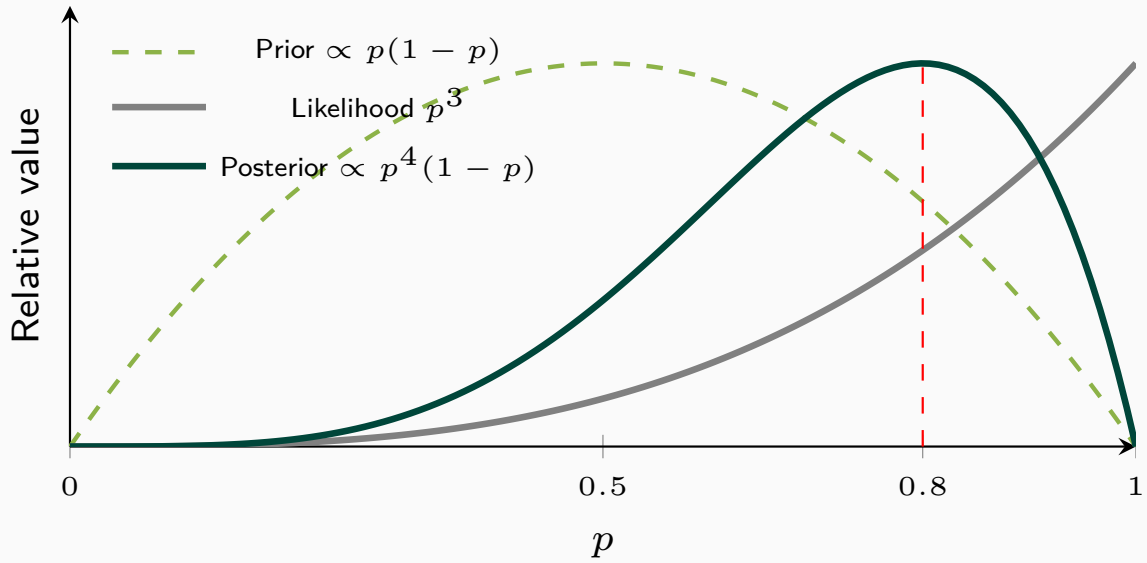
$$\widehat{p}_{\mathrm{MAP}} = \frac{k + 1}{n + 2}.$$

This is like adding **one imaginary success** and **one imaginary failure** before counting. It is also known as *Laplace smoothing*.

AI Example

In Naive Bayes text classification, Laplace smoothing prevents an unseen word from giving a probability of exactly 0.

Visualizing Prior, Likelihood, and Posterior



- The likelihood peaks at the boundary $p = 1$ (MLE).
- The posterior peaks at $p = 0.8$ (MAP): a compromise between prior and data.

Effect of Sample Size

With the prior $a = b = 2$, so that $\widehat{p}_{\mathrm{MAP}} = (k+1)/(n+2)$:

n	k	$\widehat{p}_{\mathrm{MLE}}$	$\widehat{p}_{\mathrm{MAP}}$
3	3	1.000	0.800
30	30	1.000	0.969
300	300	1.000	0.997
10	5	0.500	0.500
10	8	0.800	0.750

Key Message

- With **few data**, the prior has a strong influence.
- With **many data**, the likelihood dominates and MAP $\approx$ MLE.

From Priors to Regularization

Minimizing the **negative** log-posterior:

$$\widehat{\theta}_{\mathrm{MAP}} = \arg\min_{\theta} \Big[\underbrace{-\log p(D \mid \theta)}_{\text{data-fit loss}} + \underbrace{\big(-\log p(\theta)\big)}_{\text{penalty}} \Big].$$

General Pattern in Machine Learning

$$\boxed{\text{training objective} = \text{loss} + \lambda \times \text{regularization penalty}}$$

- The loss comes from the **likelihood** (Chapters 8–9).
- The penalty comes from the **prior**.
- A prior that prefers small weights produces a penalty on large weights.

Gaussian Prior on a Single Weight

Consider a line through the origin (Chapter 9, Slide 8):

$$y_i = wx_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2).$$

Place a Gaussian prior on the weight:

$$w \sim N(0, \tau^2), \qquad p(w) \propto \exp\left(-\frac{w^2}{2\tau^2}\right).$$

Negative log-posterior (dropping constants):

$$\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - wx_i)^2 + \frac{w^2}{2\tau^2}.$$

Multiplying by $2\sigma^2$ gives the equivalent objective:

$$\boxed{J(w) = \sum_{i=1}^{n}(y_i - wx_i)^2 + \lambda w^2, \qquad \lambda = \frac{\sigma^2}{\tau^2}}$$

Deriving the Single-Weight Ridge Estimate

$J(w)$ depends on one variable, so single-variable calculus suffices:

$$J'(w) = -2\sum_{i=1}^{n} x_i(y_i - wx_i) + 2\lambda w = 0.$$

Rearranging:

$$\sum x_i y_i = w\Big(\sum x_i^2 + \lambda\Big).$$

Result

$$\boxed{\widehat{w}_{\text{MAP}} = \frac{\sum x_i y_i}{\sum x_i^2 + \lambda}} \qquad \text{vs.} \qquad \widehat{w}_{\text{MLE}} = \frac{\sum x_i y_i}{\sum x_i^2}$$

Since $J''(w) = 2\big(\sum x_i^2 + \lambda\big) > 0$, this is a minimum.

Shrinkage

Adding $\lambda > 0$ to the denominator shrinks the estimate towards 0.

Worked Example: Shrinkage

Data: $(1, 2),\ (2, 4),\ (3, 5)$. Then:

$$\sum x_i y_i = 2 + 8 + 15 = 25, \qquad \sum x_i^2 = 1 + 4 + 9 = 14.$$

λ	$\widehat{w} = 25/(14 + \lambda)$	Comment
0	1.786	MLE / OLS
1	1.667	Mild shrinkage
6	1.250	Moderate shrinkage
36	0.500	Strong shrinkage
$\to \infty$	$\to 0$	Prior dominates

Interpretation

$\lambda = \sigma^2/\tau^2$ is large when the prior is narrow (τ^2 small), i.e. when we strongly believe the weight is close to 0.

Ridge Regression: Vector Form

With a Gaussian prior on every weight, $w_j \sim N(0, \tau^2)$ independently, the MAP objective becomes:

$$\boxed{J(\mathbf{w}) = \|\mathbf{y} - X\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2}$$

The modified normal equations and solution are:

$$(X^T X + \lambda I)\,\widehat{\mathbf{w}} = X^T \mathbf{y}, \qquad \widehat{\mathbf{w}}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T \mathbf{y}.$$

- For $\lambda > 0$, $X^T X + \lambda I$ is **always invertible**, which fixes the problem from Chapter 9, Slide 26.
- In practice, the intercept w_0 is usually **not** penalized.
- The derivation follows the same steps as Chapter 9 [Extension].

Laplace Prior and L1 Regularization

A Laplace prior has a sharp peak at 0:

$$p(w_j) \propto \exp\left(-\frac{|w_j|}{b}\right).$$

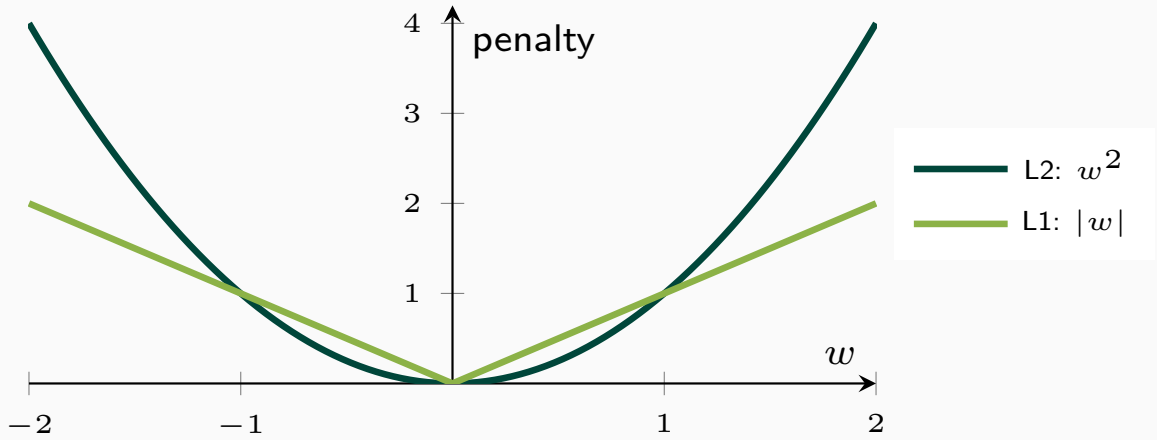
Its negative logarithm is proportional to $|w_j|$, so the MAP objective becomes:

$$\boxed{J(\mathbf{w}) = \|\mathbf{y} - X\mathbf{w}\|_2^2 + \lambda \sum_j |w_j|}$$

This is called **Lasso** regression.

- L1 regularization tends to set some weights **exactly to zero**.
- It therefore performs automatic **feature selection**.
- There is generally no closed-form solution; numerical algorithms are used.

Comparing L2 and L1 Penalties



	L2 (Ridge)	L1 (Lasso)
Prior	Gaussian	Laplace
Effect	Shrinks all weights smoothly	Sets some weights to 0
Closed form	Yes	Generally no

Near $w = 0$, the L2 penalty is very flat, whereas the L1 penalty keeps pushing small weights to 0.

Choosing the Regularization Strength λ

- λ too small: close to OLS, with a risk of **overfitting**.
- λ too large: weights shrink too much, with a risk of **underfitting**.

Standard Practice

1. Split the data into training and validation sets.
2. Fit the model for several values of λ, e.g. $0.01, 0.1, 1, 10, 100$.
3. Choose the λ with the smallest **validation** error.
4. Repeat over several splits (cross-validation) for a more stable choice.

Key Point

Regularization usually *increases* training error slightly but can *reduce* error on new data.

Summary of Part A

Prior on parameter	Penalty $-\log p(\theta)$	ML name
Uniform	Constant	No regularization (MLE)
Gaussian $N(0, \tau^2)$	$\lambda \|\mathbf{w}\|_2^2$	L2 / Ridge / weight decay
Laplace	$\lambda \|\mathbf{w}\|_1$	L1 / Lasso
Beta$(2, 2)$ for p	$-\log p - \log(1-p)$	Laplace smoothing

$$\boxed{\widehat{\theta}_{\text{MAP}} = \arg\max_{\theta} \big[\ell(\theta) + \log p(\theta)\big]}$$

Big Picture

MLE: "What explains the data best?"

MAP: "What explains the data best, given what we believed beforehand?"

Basics of Information Theory

Why Information Theory in AI?

Information theory measures **uncertainty** and **surprise** using probability.

Concept	Use in AI/ML
Entropy	Uncertainty of a prediction or a dataset
Information gain	Choosing splits in decision trees
Cross-entropy	Standard loss for classification
KL divergence	Comparing distributions (e.g. VAEs, distillation)

Logarithm Base

- $\log_2$: information measured in **bits** (used in examples).
- $\ln$: information measured in **nats** (used in ML software).

Self-Information: Measuring Surprise

Definition

The self-information of an outcome x with probability $p(x)$ is:

$$\boxed{I(x) = -\log_2 p(x)}$$

Event	$p(x)$	$I(x)$ (bits)
Certain event	1	0
Fair coin lands heads	$1/2$	1
Fair die shows 6	$1/6$	2.585
One of 8 equally likely outcomes	$1/8$	3

- Rare events carry **more** information (more surprise).
- For independent events, information adds: $I(x, y) = I(x) + I(y)$.

Entropy: Average Surprise

Definition

For a discrete random variable X with PMF $p(x)$:

$$\boxed{H(X) = E[I(X)] = -\sum_x p(x) \log_2 p(x)}$$

(with the convention $0 \log 0 = 0$).

- $H(X) \geq 0$.
- $H(X) = 0$ if and only if the outcome is certain.
- For K possible outcomes, $H(X) \leq \log_2 K$, with equality for the uniform distribution.

Interpretation

Entropy measures how unpredictable X is: the average number of bits needed to encode its outcome.

Worked Example: Computing Entropy

Example 1. $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$:

$$H = \tfrac{1}{2}\log_2 2 + \tfrac{1}{4}\log_2 4 + \tfrac{1}{4}\log_2 4 = 0.5 + 0.5 + 0.5 = 1.5 \text{ bits.}$$

Example 2. Fair die:

$$H = -\sum_{x=1}^{6} \tfrac{1}{6}\log_2 \tfrac{1}{6} = \log_2 6 \approx 2.585 \text{ bits.}$$

Example 3. A heavily biased coin, $p = (0.99, 0.01)$:

$$H \approx 0.99(0.0145) + 0.01(6.644) \approx 0.081 \text{ bits.}$$

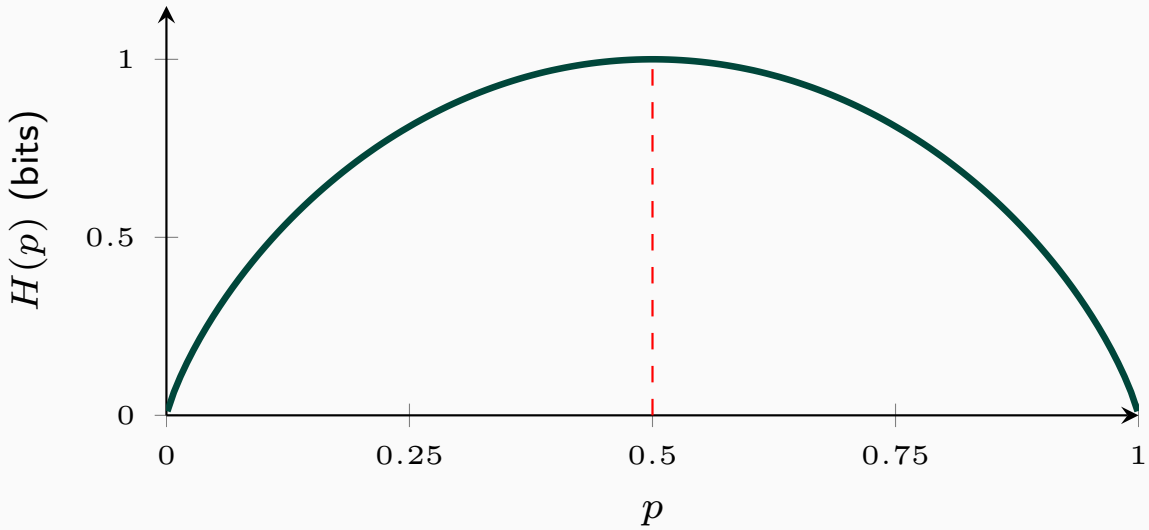
Observation

The more evenly spread the probabilities, the higher the entropy.

Binary Entropy

For a Bernoulli(p) variable:

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p).$$



- Maximum of 1 bit at $p = 0.5$, where the outcome is hardest to predict.
- $H(0) = H(1) = 0$, where the outcome is certain.

Information Gain

Suppose a feature splits a dataset S into groups $S_1, \ldots, S_m$.

Definition

$$\boxed{\mathrm{IG} = H(S) - \sum_{j=1}^{m} \frac{|S_j|}{|S|} H(S_j)}$$

Entropy before the split minus the weighted average entropy after the split.

- $H(S)$: uncertainty of the label in the whole dataset.
- The weighted sum is the remaining uncertainty after observing the feature.
- Large IG: the feature tells us a lot about the label.

AI Use

Decision tree algorithms (e.g. ID3, C4.5) choose the split with the **largest information gain**.

Worked Example: Information Gain

Eight emails: 4 spam and 4 not spam, so $H(S) = 1$ bit.

Feature A: contains the word "free"?

Group	Spam	Not spam	Entropy
Contains "free" (4 emails)	3	1	$H(0.75) \approx 0.811$
No "free" (4 emails)	1	3	$H(0.25) \approx 0.811$

$$\mathrm{IG}_A = 1 - \left(\tfrac{4}{8} \times 0.811 + \tfrac{4}{8} \times 0.811\right) \approx 0.189 \text{ bits.}$$

Feature B: a perfect split (4 spam | 4 not spam). Each group has entropy 0:

$$\mathrm{IG}_B = 1 - 0 = 1 \text{ bit.}$$

Decision

A decision tree would split on Feature B first.

Cross-Entropy

Let p be the **true** distribution and q a **model** distribution.

Definition

$$\boxed{H(p, q) = -\sum_{x} p(x) \log q(x)}$$

- Average surprise when outcomes follow p but we use the model q.
- $H(p, q) \geq H(p)$, with equality if and only if $q = p$.

Example. Three classes, true class is class 1, so $p = (1, 0, 0)$.

Model prediction q	$H(p, q) = -\ln q_1$ (nats)
$(0.7, 0.2, 0.1)$	0.357
$(0.1, 0.6, 0.3)$	2.303

Cross-Entropy as a Classification Loss

For a training example with true class y_i (one-hot p) and predicted probabilities q_i:

$$H(p, q_i) = -\log q_i(y_i).$$

Averaging over n examples:

$$\boxed{\mathcal{L}_{\mathrm{CE}} = -\frac{1}{n}\sum_{i=1}^{n} \log q_i(y_i)}$$

Connection to Chapter 8

This is exactly the **average negative log-likelihood** of a categorical model. For two classes, it reduces to the binary cross-entropy:

$$-\frac{1}{n}\sum_{i=1}^{n}\big[y_i \log p_i + (1-y_i)\log(1-p_i)\big].$$

Minimizing cross-entropy = maximizing likelihood.

Kullback–Leibler (KL) Divergence

Definition

$$D_{\mathrm{KL}}(p \,\|\, q) = \sum_{x} p(x) \log \frac{p(x)}{q(x)}$$

- $D_{\mathrm{KL}}(p\|q) \geq 0$ (Gibbs' inequality).
- $D_{\mathrm{KL}}(p\|q) = 0$ if and only if $p = q$.
- **Not symmetric**: in general $D_{\mathrm{KL}}(p\|q) \neq D_{\mathrm{KL}}(q\|p)$.
- Requires $q(x) > 0$ wherever $p(x) > 0$; otherwise it is infinite.

Interpretation

KL divergence is the **extra** average surprise caused by using q instead of the true distribution p. It measures how different q is from p, but it is not a true distance.

Worked Example: KL Divergence Is Not Symmetric

Let $p = (\frac{1}{2}, \frac{1}{2})$ and $q = (\frac{3}{4}, \frac{1}{4})$. Using $\log_2$:

$$D_{\mathrm{KL}}(p\|q) = \tfrac{1}{2}\log_2 \frac{1/2}{3/4} + \tfrac{1}{2}\log_2 \frac{1/2}{1/4} \approx \tfrac{1}{2}(-0.585) + \tfrac{1}{2}(1) = 0.208.$$

$$D_{\mathrm{KL}}(q\|p) = \tfrac{3}{4}\log_2 \frac{3/4}{1/2} + \tfrac{1}{4}\log_2 \frac{1/4}{1/2} \approx \tfrac{3}{4}(0.585) + \tfrac{1}{4}(-1) = 0.189.$$

Conclusion

$0.208 \neq 0.189$: the order of the two distributions matters.

Cross-Entropy = Entropy + KL Divergence

Since $\log \frac{p}{q} = \log p - \log q$:

$$\boxed{H(p, q) = H(p) + D_{\mathrm{KL}}(p \,\|\, q)}$$

Check with $p = (\frac{1}{2}, \frac{1}{2})$, $q = (\frac{3}{4}, \frac{1}{4})$:

$$H(p) = 1, \qquad H(p, q) = \tfrac{1}{2}(0.415) + \tfrac{1}{2}(2) \approx 1.208,$$

$$H(p) + D_{\mathrm{KL}}(p\|q) \approx 1 + 0.208 = 1.208. \checkmark$$

Why This Matters in Training

The true distribution p is fixed, so $H(p)$ is a constant. Therefore:

$$\text{minimizing cross-entropy} \iff \text{minimizing } D_{\mathrm{KL}}(p\|q).$$

Further Topics in Information Theory [Extension]

- **Joint entropy**: $H(X,Y) = -\sum_{x,y} p(x,y) \log p(x,y)$
- **Conditional entropy**: $H(Y \mid X) = H(X,Y) - H(X)$
- **Mutual information**:
  $I(X;Y) = H(Y) - H(Y \mid X) = D_{\mathrm{KL}}\big(p(x,y) \,\|\, p(x)p(y)\big)$
- Information gain in decision trees is the mutual information between the feature and the label.
- Proof of Gibbs' inequality $D_{\mathrm{KL}} \geq 0$ uses $\ln t \leq t - 1$.

Not Examined

These topics are provided for interested students and future courses in machine learning.

Introduction to Markov Chains (Not Examined)

What Is a Markov Chain? [Not Examined]

A sequence of random states $X_0, X_1, X_2, \ldots$ taking values in a finite set of **states**.

Markov Property

The future depends only on the **present**, not on the past:

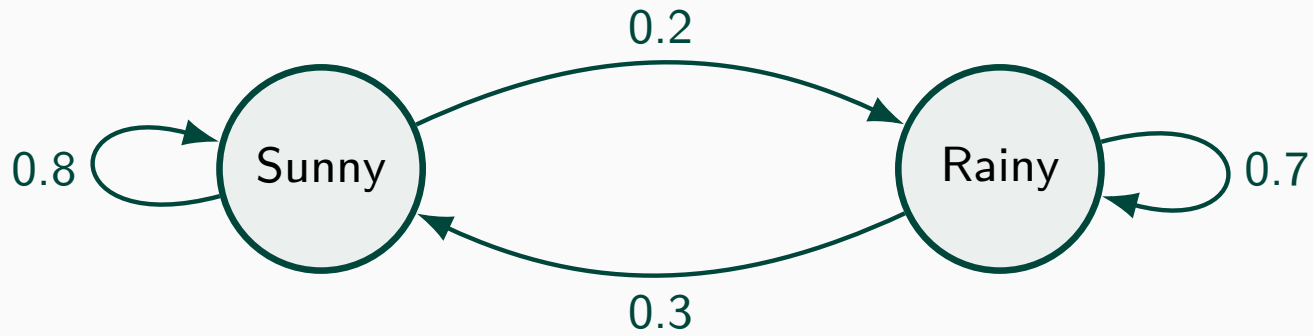
$$P(X_{t+1} = j \mid X_t = i, X_{t-1}, \ldots, X_0) = P(X_{t+1} = j \mid X_t = i).$$

- **Transition probability**: $P_{ij} = P(X_{t+1} = j \mid X_t = i)$.
- Examples: weather (sunny/rainy), user behaviour on a website, words in a sentence.

Connection to Week 2

A Markov Chain is a repeated application of conditional probability and the law of total probability.

Transition Diagram and Transition Matrix [Not Examined]



$$P = \begin{matrix} \\ \text{from Sunny} \\ \text{from Rainy} \end{matrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} \qquad \text{(columns: to Sunny, to Rainy)}$$

- Every entry satisfies $0 \leq P_{ij} \leq 1$.
- Every **row** sums to 1 (row-stochastic matrix).

Updating the State Distribution [Not Examined]

Let $\boldsymbol{\pi}_t$ be a **row vector** of state probabilities at time t.

By the law of total probability:

$$\boxed{\boldsymbol{\pi}_{t+1} = \boldsymbol{\pi}_t P}$$

Example. Today is sunny: $\boldsymbol{\pi}_0 = \begin{bmatrix} 1 & 0 \end{bmatrix}$.

$$\boldsymbol{\pi}_1 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} = \begin{bmatrix} 0.8 & 0.2 \end{bmatrix},$$

$$\boldsymbol{\pi}_2 = \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} P = \begin{bmatrix} 0.64 + 0.06 & 0.16 + 0.14 \end{bmatrix} = \begin{bmatrix} 0.70 & 0.30 \end{bmatrix}.$$

Linear Algebra Connection

Predicting the future of a Markov Chain is repeated vector–matrix multiplication.

Long-Run Behaviour and Applications [Extension]

t	0	1	2	3	$\cdots$	∞
$P(\text{Sunny})$	1.00	0.80	0.70	0.65	$\cdots$	0.60
$P(\text{Rainy})$	0.00	0.20	0.30	0.35	$\cdots$	0.40

A **stationary distribution** satisfies $\boldsymbol{\pi} = \boldsymbol{\pi} P$. Check:

$$\begin{bmatrix} 0.6 & 0.4 \end{bmatrix} P = \begin{bmatrix} 0.48 + 0.12 & 0.12 + 0.28 \end{bmatrix} = \begin{bmatrix} 0.6 & 0.4 \end{bmatrix}.$$

AI Applications

- **PageRank**: stationary distribution of a random web surfer
- n-**gram language models**: next word depends on recent words
- **Reinforcement learning**: Markov decision processes

Classwork: A Three-State Markov Chain

A website user is in one of three states: **B**rowse, **C**art, or **P**urchase.

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.5 & 0.1 & 0.4 \end{bmatrix} \qquad \text{(rows and columns ordered B, C, P)}$$

Tasks

1. Verify that P is a valid transition matrix.
2. The user starts in Browse. Write the initial state vector $\boldsymbol{\pi}_0$.
3. Compute $\boldsymbol{\pi}_1 = \boldsymbol{\pi}_0 P$.
4. Compute $\boldsymbol{\pi}_2 = \boldsymbol{\pi}_1 P$. What is the probability that the user is in Purchase after two steps?

Assessment Note

This classwork counts towards A1 (ILO M5). Markov Chains are not examined in the Test or Examination.

Classwork: Solution

1. All entries lie in $[0, 1]$, and each row sums to 1:
   $0.6 + 0.3 + 0.1 = 0.2 + 0.5 + 0.3 = 0.5 + 0.1 + 0.4 = 1.$
2. $\boldsymbol{\pi}_0 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$.
3. $\boldsymbol{\pi}_1 = \begin{bmatrix} 0.6 & 0.3 & 0.1 \end{bmatrix}$ (the first row of P).
4. $$\boldsymbol{\pi}_2 = \begin{bmatrix} 0.36 + 0.06 + 0.05 & 0.18 + 0.15 + 0.01 & 0.06 + 0.09 + 0.04 \end{bmatrix}$$
   $$= \begin{bmatrix} 0.47 & 0.34 & 0.19 \end{bmatrix}.$$

Answer

$P(\text{Purchase after two steps}) = 0.19$. Check: $0.47 + 0.34 + 0.19 = 1$.

Summary and Exercises

Summary: Key Definitions and Formulas

$$\widehat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \big[\log p(D \mid \theta) + \log p(\theta)\big]$$

Bernoulli: $\widehat{p}_{\mathrm{MAP}} = \dfrac{k + a - 1}{n + a + b - 2}$ Ridge: $\widehat{\mathbf{w}} = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$

$$H(X) = -\sum_{x} p(x) \log p(x) \qquad \mathrm{IG} = H(S) - \sum_{j} \frac{|S_j|}{|S|} H(S_j)$$

$$H(p, q) = -\sum_{x} p(x) \log q(x) \qquad D_{\mathrm{KL}}(p \| q) = \sum_{x} p(x) \log \frac{p(x)}{q(x)}$$

$$H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$$

Markov Chain [Not Examined]: $\boldsymbol{\pi}_{t+1} = \boldsymbol{\pi}_t P$

Common Mistakes

- Thinking that MAP gives the full posterior distribution
- Forgetting that MAP equals MLE under a uniform prior
- Believing that a larger λ always gives a better model
- Penalizing the intercept term in ridge regression without reason
- Mixing $\log_2$ and $\ln$ within the same calculation
- Writing entropy without the minus sign, giving a negative value
- Treating KL divergence as symmetric or as a distance
- Forgetting that $\log q(x)$ is undefined when $q(x) = 0$
- Multiplying $P\boldsymbol{\pi}_t$ instead of $\boldsymbol{\pi}_t P$ for a row vector

Class Exercise

Problem 1: Bernoulli MAP

A coin is flipped 4 times and lands heads every time. With prior $a = b = 2$, compute $\widehat{p}_{\mathrm{MLE}}$ and $\widehat{p}_{\mathrm{MAP}}$.

Problem 2: Shrinkage

For data $(1, 2)$, $(2, 3)$ and the model $y = wx + \varepsilon$, compute $\widehat{w}_{\mathrm{MLE}}$ and the ridge estimate with $\lambda = 3$.

Problem 3: Entropy and Cross-Entropy

1. Compute $H(p)$ in bits for $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$.
2. A classifier predicts $q = (0.2, 0.5, 0.3)$ and the true class is class 2. Compute the cross-entropy loss in nats.

Work in pairs for 15 minutes, then discuss as a class.

Class Exercise: Solutions to Problems 1 and 2

Problem 1. $n = 4$, $k = 4$:

$$\widehat{p}_{\text{MLE}} = \frac{4}{4} = 1, \qquad \boxed{\widehat{p}_{\text{MAP}} = \frac{4+1}{4+2} = \frac{5}{6} \approx 0.833}$$

Problem 2.

$$\sum x_i y_i = 1 \times 2 + 2 \times 3 = 8, \qquad \sum x_i^2 = 1 + 4 = 5.$$

$$\widehat{w}_{\text{MLE}} = \frac{8}{5} = 1.6, \qquad \boxed{\widehat{w}_{\text{ridge}} = \frac{8}{5+3} = 1.0}$$

Interpretation

Both priors pull the estimate away from the extreme or unregularized value.

Class Exercise: Solution to Problem 3

Part 1.

$$H(p) = \tfrac{1}{2}\log_2 2 + \tfrac{1}{4}\log_2 4 + \tfrac{1}{8}\log_2 8 + \tfrac{1}{8}\log_2 8$$
$$= 0.5 + 0.5 + 0.375 + 0.375 = \boxed{1.75 \text{ bits}}$$

Part 2. The true distribution is one-hot $p = (0, 1, 0)$:

$$H(p, q) = -\ln q_2 = -\ln 0.5 \approx \boxed{0.693 \text{ nats}}$$

Interpretation

If the model had predicted $q_2 = 0.9$, the loss would fall to $-\ln 0.9 \approx 0.105$. Confident correct predictions give small loss.

Preview of Next Week: Confidence Intervals

- So far, our estimates have been **single numbers** (point estimates): MLE, OLS, MAP.
- But how **uncertain** are these estimates?
- Next week, we will use the sampling distribution and the CLT (Week 7) to build **interval estimates**:

$$\overline{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \qquad \overline{x} \pm t_{\alpha/2,\, n-1}\frac{s}{\sqrt{n}}.$$

- We will also learn how to interpret a 95% confidence interval correctly.

Main Connection

A point estimate tells us our best guess; a confidence interval tells us how much to trust it.